# Lab 5

## 1. Import the libraries

In [2]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

print("OpenCV version:", cv2.__version__)
```

```
OpenCV version: 5.0.0
```

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## 2. Read and prepare the image

In [3]:
```
image_path = Path("images/Parrot.png")  # Change this path if your image is

image_bgr = cv2.imread(str(image_path))
if image_bgr is None:
    raise FileNotFoundError(
        f"Could not load image: {image_path.resolve()}\n"
        "Place an image at this path or update image_path to your image file
    )

image = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(7, 5))
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")
plt.show()

print("Image shape:", image.shape)
```




```
Image shape: (554, 554, 3)
```

---

# Task 1:

In [4]:
```
box_smoothed = cv2.blur(image, (7, 7))

fig, ax = plt.subplots(1, 2, figsize=(12, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(box_smoothed)
ax[1].set_title("Smoothed Image (7×7 Box Filter)")
ax[1].axis("off")

plt.tight_layout()
plt.show()
```




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# Task 2:

In [5]:
```
gaussian_5 = cv2.GaussianBlur(image, (5, 5), 0)
gaussian_21 = cv2.GaussianBlur(image, (21, 21), 0)

fig, ax = plt.subplots(1, 3, figsize=(15, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(gaussian_5)
ax[1].set_title("Gaussian Filter (5×5)")
ax[1].axis("off")

ax[2].imshow(gaussian_21)
ax[2].set_title("Gaussian Filter (21×21)")
ax[2].axis("off")

plt.tight_layout()
plt.show()
```



# Task 3:

In [7]:
```
image_float = image.astype(np.float32) / 255.0

laplacian = cv2.Laplacian(image_float, cv2.CV_32F, ksize=3)

sharpened = np.clip(image_float - laplacian, 0, 1)

laplacian_display = np.abs(laplacian)
max_value = laplacian_display.max()
if max_value > 0:
    laplacian_display = laplacian_display / max_value

fig, ax = plt.subplots(1, 3, figsize=(15, 5))
ax[0].imshow(image_float)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(laplacian_display)
ax[1].set_title("Laplacian Response (Absolute, Normalized)")
ax[1].axis("off")

ax[2].imshow(sharpened)
ax[2].set_title("Sharpened Image")
ax[2].axis("off")

plt.tight_layout()
plt.show()
```

